

# Assignment 1: Financial Engineering

## Numerical Pricing of Vanilla and Barrier Options

### Group 12

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### Abstract

This report provides a detailed, step-by-step analysis of Assignment 1. The document is structured to strictly follow the assignment's prompts, outlining the numerical results, methodological choices, and, most importantly, the logical and computational criticalities identified and resolved during implementation.

## Initial Parameters

| Parameter                  |                                     |
|----------------------------|-------------------------------------|
| Underlying                 | Equity stock                        |
| Underlying price ( $S_0$ ) | 1.00 Euro                           |
| Strike price ( $K$ )       | 1.10 Euro                           |
| Value date                 | 15 <sup>th</sup> of February 2008   |
| Time-to-maturity (TTM)     | 4 months (year fraction $T = 1/3$ ) |
| Volatility ( $\sigma$ )    | 21.2% per year                      |
| Risk-free rate ( $r$ )     | 2.5%                                |
| Dividend Yield ( $d$ )     | 2.0%                                |
| Settlement                 | Physical delivery                   |
| Number of contracts        | 1 Million                           |

## Point A: Pricing the European Call Option

The first task required pricing a European Call option using three distinct methods: the Black-76 closed-form formula, the Cox-Ross-Rubinstein (CRR) binomial tree, and Monte Carlo (MC) model. Choosing  $M = 100$ , we obtained the following results:

- **Closed Formula (Black-76):**  $\approx 0.0163$  €
- **CRR Model :**  $\approx 0.0163$  €
- **Monte Carlo :**  $\approx 0.0150$  €

The Monte Carlo simulation yields a significantly different result, as  $M = 100$  iterations are insufficient to achieve an acceptable level of precision.

Multiplying the price by the derivative notional of 1 Million, the total value amounts to approximately **16264 €**.

### Functions

```
[OptionPrice] = EuropeanOptionPrice(F0, K, B, T, sigma, pricingMode, M, flag);
[OptionPrice] = EuropeanOptionClosed(F0,K,B,T,sigma,flag);
[OptionPrice] = EuropeanOptionCRR(F0,K,B,T,sigma,N,flag);
[OptionPrice] = EuropeanOptionMC(F0,K,B,T,sigma,N,flag);
```

## Point B: Selection of Parameter $M$

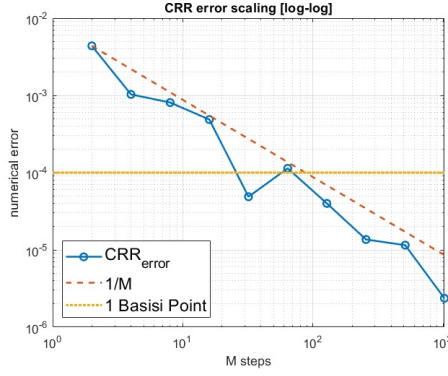
The criterion used to select the number of time steps (CRR) and simulations (MC) is based on the market bid/ask spread for equity options, specified as 1 basis point (0.01%). Given a Spot price  $S_0 = 1$  Euro, the absolute error tolerance was strictly set to less than **0.0001 Euro**.

- **For the CRR model:** Since CRR is a discrete-time model and each step can either surpass the strike or not depending on the single jump, it is possible to find a value of  $M$  such that the error is smaller than the threshold of 1 bp, while for some  $M' > M$  the error may increase again. To ensure robustness in the choice of  $M$ , we implemented a *while* loop that verifies not only the current iteration but also the subsequent ones. Specifically, the algorithm checks a forward-looking window of 50 consecutive values of  $M$ . The loop continues until all errors in this window remain below the precision level of  $10^{-4}$  Euro. The resulting stable choice is  **$M = 86$** .
- **For the MC model:** The standard error decays slowly, at rate  $\mathcal{O}\left(\frac{1}{\sqrt{M}}\right)$ . To reach the desired accuracy, instead of performing each simulation sequentially up to approximately 200,000, we increased  $M$  in steps of 1,000, implying that the precision of our search procedure is of the order of thousands. The selected value is  **$M = 207,000$** .

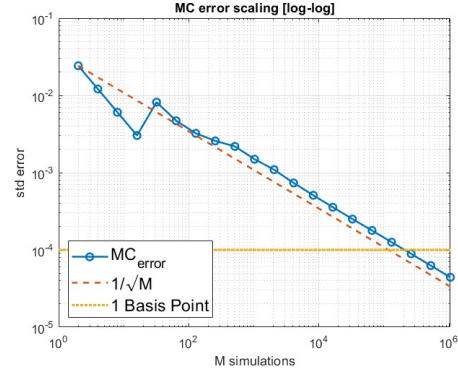
## Point C: Error Analysis (Log-Log Scale)

We plotted the numerical errors of both numerical methods against varying values of  $M$  on a logarithmic scale.

- **For the CRR model:** The CRR plot highlights a downward trend parallel to the theoretical  $\mathcal{O}\left(\frac{1}{M}\right)$  line, while clearly exhibiting significant oscillations (the so-called “*sawtooth effect*”). As the number of steps  $M$  increases, the nodes of the tree shift relative to the strike price  $K$ , causing the error to fluctuate while maintaining a downward trend. The oscillations persist as  $M$  increases, but their amplitude decreases, leading to progressively more stable estimates.
- **For the MC model:** The Monte Carlo plot displays a linear decay (in log-log scale) that closely tracks the  $\mathcal{O}\left(\frac{1}{\sqrt{M}}\right)$  asymptote, consistently with the Central Limit Theorem.



Cox-Ross-Rubinstein Error



Monte Carlo Error

**Resolved Criticality (Log-Log Plot Visual Bug):** For extremely low initial values of  $M$  (e.g.,  $M = 2$  or  $4$ ), the Monte Carlo method frequently generated trajectories that entirely finished out-of-the-money, yielding an estimated standard error of exactly zero.

### Functions

```
[M, stdEstim] = PlotErrorMC(F0, K, B, T, sigma);
[M, errorCRR] = PlotErrorCRR(F0, K, B, T, sigma);
```

## Point D: Pricing Call Option with European Barrier (Up-and-Out)

Consider a European Call Option with a knock-out level of  $KO = 1.4$  Euro. Being a *European* barrier, the algorithm applies the payoff truncation exclusively at maturity (evaluating the terminal nodes  $F_T$ ).

As expected, the calculated price is strictly lower than that of the vanilla Call, since the theoretically unlimited upside potential is capped by the barrier, eliminating the most profitable scenarios.

We also confirmed the existence of a **closed-formula solution** for this derivative. The payoff profile of the derivative can be statically replicated by going long a European Vanilla Call with strike  $K$ , short a European Vanilla Call with strike  $KO$  (bull call-spread at  $(K, KO)$ ), and short  $(KO - K)$  Digital Cash-or-Nothing options struck at  $KO$ .

$$\text{Barrier Option} = \text{Bull-Call-Spread}(K, KO) - (KO - K) \cdot \mathbb{I}_{\{F_T > KO\}}$$

### Functions

```
[optionPrice] = EuropeanOptionKOMC(F0, K, KO, B, T, sigma, N)
[optionPrice] = EuropeanOptionKOCRR(F0, K, KO, B, T, sigma, N)
```

## Point E: Sensitivity Analysis (Delta and Vega of the KO Option)

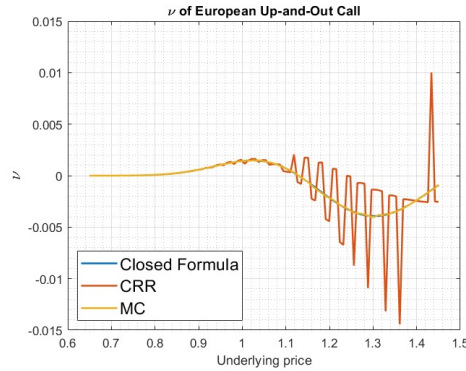
The function implemented returns the absolute price change resulting from a 1% bump in the volatility, which is standard market convention.

We compute it with both numerical approaches (CRR and MC, estimating Vega by a central finite difference method) and closed-form analytical approach (applying the linearity property of the Greeks).

## Explanation of the VegaK0 Function

The `VegaK0` function is designed to compute the volatility sensitivity (Vega) of an option portfolio that replicates a barrier payoff or a capped bull spread. The computation is carried out through three distinct methodological approaches, which can be selected using the `flagNum` parameter:

- **Finite Difference Approximation (CRR and MC Methods):** When `flagNum` is set to 1 (Cox-Ross-Rubinstein Binomial Trees) or 2 (Monte Carlo Simulation), the code estimates the derivative by computing the option price twice: once by shifting the volatility up by a small step  $\Delta\sigma$  (`dsigma = 0.01`) and once by shifting it down by the same amount. In the Monte Carlo approach, the random seed is fixed (`rng(1)`) before each evaluation; this is a crucial practice to ensure that the price difference is solely driven by the volatility shift and not by the random noise of different simulated paths.
- **Exact Analytical Solution:** When `flagNum` is set to 3, the code uses a static replication of the payoff rather than numerical approximation. The overall Vega is calculated as a linear combination of the Black-Scholes closed-form formulas for three components: a European Call with strike  $K$ , a European Call with strike  $KO$ , and a Digital Call weighted by the strike difference ( $KO - K$ ).



Comparison between Vega

When the underlying price is significantly below the strike, an increase in volatility gives the asset a better chance of finishing in-the-money without surpassing the barrier. As the underlying price moves past the strike and gets closer to the barrier, the risk of hitting the threshold becomes relevant. Hence Vega becomes negative.

The sawtooth instability in the CRR model's Vega occurs because its discrete tree nodes cannot perfectly align with a continuous barrier. Calculating Vega requires slightly perturbing the volatility ( $\sigma$ ), which consequently shifts the entire grid of nodes. When the initial price ( $S_0$ ) is close to the barrier, this minor shift can push a heavily weighted node across the boundary, causing its payoff to abruptly drop to zero. This sharp discontinuity makes the finite difference calculation spike erratically. Conversely, for lower  $S_0$  values, the probability of reaching these shifting boundary nodes is too small to significantly disrupt the pricing.

## Functions

`[vega] = VegaK0(F0, K, KO, B, T, sigma, N, flagNum)`

| Option                         | Vega Formula ( $\nu = \frac{\partial V}{\partial \sigma}$ ) |
|--------------------------------|-------------------------------------------------------------|
| European Call                  | $S\sqrt{T} \cdot n(d_1)$                                    |
| Digital Call (Cash-or-Nothing) | $-e^{-rT} n(d_2) \frac{d_1}{\sigma}$                        |

## Point F: Call Option with American Barrier

To price the Call Option with an American Up-and-Out barrier (continuous monitoring), we implemented a closed-form solution based on the reflection principle. The value of option is expressed by the following formula:

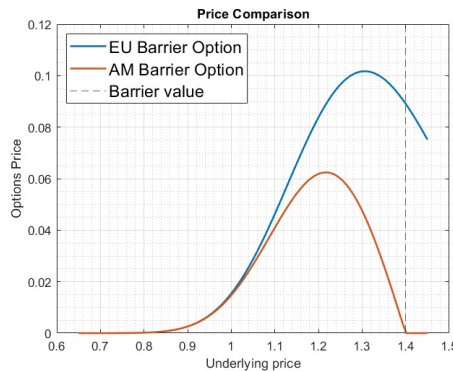
$$F^{UO}(t, s, \Phi) = F(t, s, \Phi^L) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} F\left(t, \frac{L^2}{s}, \Phi^L\right)$$

where both parts are evaluated using the European Up-and-Out barrier formula (which represents exactly the chopped payoff). The first component is the European barrier option evaluated at the standard forward price ( $F_0$ ), while the second component evaluates the same European barrier option at a reflected forward price ( $F_1$ , derived from the reflected spot  $H^2/S_0$ ). Obviously, if the initial spot price  $S_0$  is already above the barrier, the formula immediately returns zero.

### Comparison between prices

The primary observation is that the American barrier (AM) price is consistently lower than the European barrier (EUR) price across all values of  $S_0$ . This discrepancy arises from the monitoring frequency of the barrier condition:

- **American Barrier (Continuous Monitoring):** The knock-out condition is “path-dependent.” If the underlying price  $S_t$  touches or exceeds the 1.4 threshold at any time  $t$  during the 4-month life of the option, the option expires worthless immediately so the probability of hitting the barrier increasing.
- **European Barrier (At-Maturity Monitoring):** The knock-out condition is only verified at the expiration date  $T$ . Even if the stock price trades above 1.4 during the life of the contract, the option remains “alive” as long as the price settles back below 1.4 by the value date.

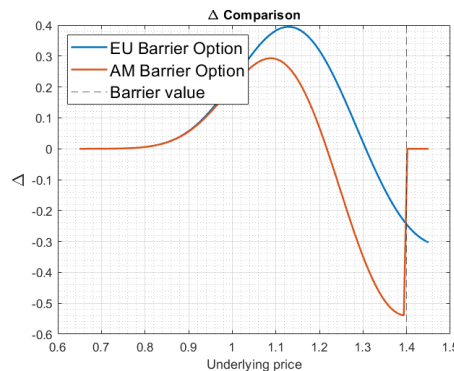


Price

## 1. Delta ( $\Delta$ ) Analysis

Delta measures the sensitivity of the option's price to small changes in the underlying asset's price.

- **Far from the Barrier** ( $S_0 < 1.0$ ): In this region, both options behave similarly to a standard Vanilla Call. As  $S_0$  increases, the option gets closer to being In-the-Money, so Delta is positive and increasing. The two curves overlap almost perfectly.
- **Approaching the Barrier** ( $1.1 < S_0 < 1.4$ ): Here we see the “barrier effect.” Instead of converging to 1.0 like a standard Call, the Delta turns strictly negative. This fact appears because, as  $S_0$  gets closer to 1.4, an increase in the underlying price actually hurts the value of the option by drastically increasing the probability of a knock-out.
- **American vs. European Dynamics:**
  - \* The **American Delta** (red line) drops into negative territory much faster and exhibits a severe negative spike right before 1.4. This reflects the extreme localized risk: if you are at 1.39, even a tiny upward tick will instantly kill the option. At exactly  $S_0 = 1.4$ , the American Delta abruptly snaps to 0 because the option ceases to exist.
  - \* The **European Delta** (blue line) also turns negative but follows a smoother trajectory. Since the barrier is only checked at maturity  $T$ , there is less localized panic; a breach today doesn't immediately kill the trade, smoothing out the sensitivity.



Delta

## 2. Vega ( $\nu$ ) Analysis

Vega measures the sensitivity of the option's price to changes in implied volatility.

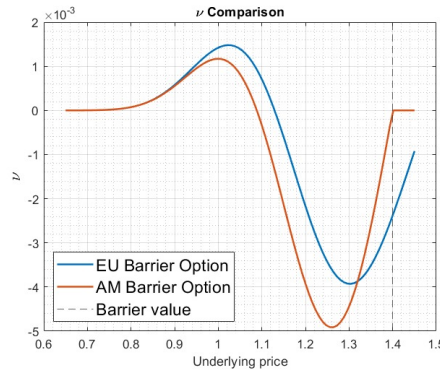
- **The Dual Nature of Volatility:**

Higher volatility increases the probability of the underlying asset's price going up, which is good for a Call option. However, it simultaneously increases the risk of the price hitting the upper barrier. If the barrier is touched, the option immediately expires worthless.
- **Positive Vega region** ( $S_0 \approx K$ ): When  $S_0$  is around the strike (1.1), higher volatility helps the option get In-the-Money without posing an immediate threat to the distant barrier.

- **Negative Vega region ( $S_0 > 1.15$ ):** As  $S_0$  approaches the barrier, higher volatility becomes the enemy. Increased volatility means wider price swings, which drastically increases the probability of hitting the 1.4 knock-out level. Consequently, Vega becomes deeply negative.

### The Difference in Volatility Risk:

- **American Options (Checked constantly):** Because the barrier is monitored all the time, any sudden price swing can instantly knock out the option. This makes it highly vulnerable to high volatility, causing its value to drop sharply if the market gets too chaotic.
- **European Options (Checked only at the end):** These are safer from daily market swings because the barrier is only checked at expiration. Wild price movements during the option's life don't trigger the knock-out, as long as the price is safe on the final day. Therefore, high volatility hurts its value much less.



Vega

### New function implemented in the code:

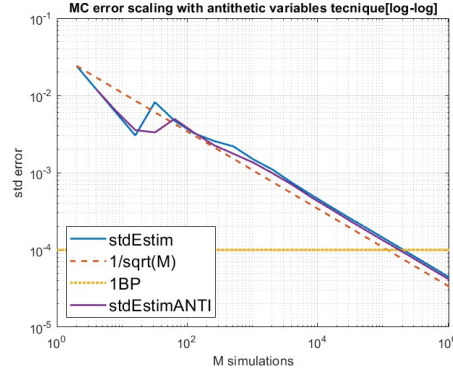
```
[optionPrice]=EuropeanOptionAmericanKOClosed(F0,S0,K, K0,B,TTM,sigma,r,d)
```

**Point G (Antithetic Variables):** We implemented the Antithetic variable technique for variance reduction. For a target number of simulations  $M$ , the algorithm generates  $M/2$  **independent standard normal draws** ( $Z$ ) and symmetrically constructs their antithetic counterparts ( $-Z$ ).

The option payoffs from these **perfectly negatively correlated** paths are then computed and averaged to form  $M/2$  independent paired estimators. By computing the standard error based on these paired averages rather than individual paths, the method demonstrates a significant **reduction** in the estimator's **variance**, improving convergence while effectively halving the required random number generations.

This approach indeed introduced bias, but lowered the Monte Carlo standard error.

The numerical results, considering the same option of point B, highlight the true power of this method. For our specific case, the standard error actually decreased from **9.9934e-05** (*standard MC*) to **9.3045e-05** (*antithetic MC*). We must stress that this improved precision was achieved while generating only half the number of random simulations.



**New function implemented in the code:**

```
[M,stdEstim]=PlotErrorANTIMC(F0,K,B,T,sigma)
```

**Points H & I (Bermudan Option and Dividends):** The function prices a Bermudan Call option using CRR binomial tree. The algorithm computes the **continuation value** at each step using risk-neutral probabilities. On predefined early exercise dates, it applies the Bermudan condition by taking the **maximum** between the continuation value and the immediate exercise payoff. Finally, the expected value at inception is discounted to the present day to return the price.

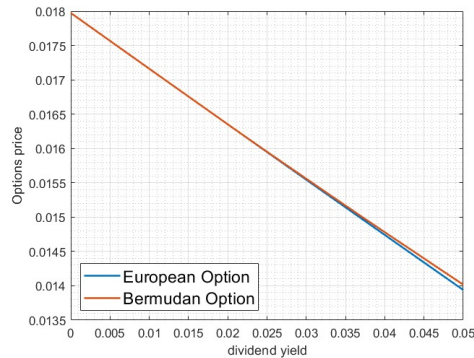
The early exercise of a Bermudan call option can be optimal when the underlying asset pays dividends. A high dividend yield reduces the cost of carry, which decreases the forward price relative to the spot price according to the formula

$$F_t = S_t \times e^{(r-q)(T-t)}$$

In particular, for a stock that pays no dividends (or very small dividends), it is **never optimal to exercise** a Call option early, since you immediately capture the intrinsic value, but you throw away the option's remaining time value. Because the Bermudan option's early exercise feature is effectively useless at low dividend yields, its price perfectly matches the standard European option.

As the dividend yield grows, the **expected drop** in the stock price becomes **significant**, so we conclude that the critical value of dividend yield given our setting is  $\delta^* \approx 5\%$ , as the value of the European and Bermudan barriers start to diverge, with Bermudan price greater than the European one, as expected. This occurs because a Bermudan exercise option sits between its European and American counterparts, following the relation:

$$\text{European} \leq \text{Bermudan} \leq \text{American}$$



### New function implemented in the code:

Bermudan\_price=BermudanOptionCRR2(F0,K,T,B,sigma,r,q,M)

### Comparison between ATM Forward Call and Put Options:

The relationship between European Call and Put options is governed by the Put-Call parity. For options where the underlying is a forward contract (or adjusting for the dividend yield in Black-Scholes), the parity is expressed as:

$$C - P = e^{-rT}(F - K)$$

By definition, for an At-The-Money (ATM) Forward option, the strike price is exactly equal to the current forward price ( $K = F$ ). Consequently, the right-hand side of the equation reduces to zero, leading to:

$$C = P$$

This mathematical equivalence demonstrates that ATM Forward Calls and Puts have identical premiums.

While the prices are indistinguishable, the fundamental difference between the two instruments lies in their directional exposure to the underlying, measured by the option's Delta ( $\Delta$ ). For an ATM Forward option, where  $d_1 = \frac{\sigma\sqrt{T}}{2}$ , the probabilities approach 0.5. Therefore, the deltas exhibit opposite signs but equal magnitude:

$$\Delta_{\text{ATM Call}} = 0.5 \quad \text{and} \quad \Delta_{\text{ATM Put}} = -0.5$$

Hence,  $\Delta_{\text{ATM Call}} = -\Delta_{\text{ATM Put}}$ .

**What if the Forward can take negative values:** The choice of a GBM model is no longer valid, as it ensures that the underlying is always positive. Therefore, under risk neutral measure we have

$$dS_t = (r - q)S_t dt + \sigma dW_t^{\mathbb{Q}}$$

Using the relationship  $F_t = S_t e^{(r-q)(T-t)}$  and applying Ito, we get

$$dF_t = \sigma e^{(r-q)(T-t)} dW_t^{\mathbb{Q}}$$

$$C = e^{-rT} E^{\mathbb{Q}}[\max(F_T - K, 0)]$$

Where  $F_t \sim N(0, \Sigma^2)$  and  $\Sigma^2 = \int \sigma_F^2 ds$ , and  $\sigma_F = \sigma e^{r(T-t)}$

Therefore

$$C = e^{-r(T-t)} \left[ (F_0 - K) \Phi\left(\frac{F_0 - K}{\Sigma}\right) + \Sigma \phi\left(\frac{F_0 - K}{\Sigma}\right) \right]$$

with

- $\Phi(x)$ : The cumulative distribution function of the standard normal.
- $\phi(x)$ : The probability density function of the standard normal.
- $\Sigma$ : The volatility of the underlying Forward.

## Challenges

- Selection of a sufficiently large number of steps (in the CRR model) or simulations (in the MC model) in order to satisfy the tolerance condition of 1 bp. In principle, one could increase  $M$  until the error strictly meets the prescribed accuracy. However, such a brute-force approach is not computationally feasible, especially for the Monte Carlo framework, where convergence is slow and the computational cost grows linearly with the number of simulations.
- Since the logarithm of zero is undefined, MATLAB's `loglog` function halted the graphical rendering. We resolved this by ignoring the results for low  $M$ , as very low monte carlo sample estimates are inherently noisy.
- The GBM model will break if we ever get a value of  $S_t < 0$